

Automatic Guided Vehicle – AGV

The AGV can be ordered to move to five predefined positions (A, B, C, D and HOME). All motion programs are already downloaded in the AGV. Send an order to the AGV and then wait for an answer from the AGV after it has stopped. The communication between PLC and AGV must follow the protocol (handshake) described below. If the protocol is not followed, or a wrong order is given, the AGV will go into error state (jump away) and must be repaired (restart simulation).

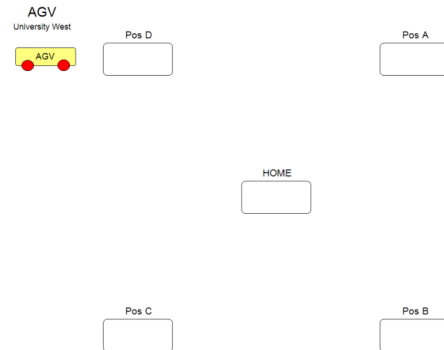
I/O definitions for AGV virtual process

//Input to PLC from process

ixAGVStrobe: BOOL; (* AGV strobe signal *)
iiAtPos: INT; (* Actual position of AGV *)

//Output from PLC to process

qxPLCStrobe: BOOL; (* PLC strobe signal *)
qiGotoPos: INT; (* Target position for AGV *)



Messages

Order to AGV

- 1 Go to position HOME
- 2 Go to position A
- 3 Go to position B
- 4 Go to position C
- 5 Go to position D
- 6 Stop instantly

Answer from AGV

- 1 At position HOME
- 2 At position A
- 3 At position B
- 4 At position C
- 5 At position D
- 6 Not at any defined position

Handshake PLC – AGV

- a) AGV read order from PLC at rising edge of PLC strobe.
- b) AGV send 'order received' (answer = 0 and rising edge of AGV strobe).
- c) AGV wait for PLC 'send ready' (falling edge of PLC strobe).
- d) AGV send 'ack strobe' (falling edge of AGV strobe) and start moving.
- e) AGV send 'answer ready' (answer > 0 and rising edge of AGV strobe) after AGV stopped.
- f) AGV wait for PLC 'answer received' (order = 0 and rising edge of PLC strobe).
- g) AGV send 'answer end' (answer = 0 and falling edge of AGV strobe).
- h) AGV wait for PLC 'ack strobe' (falling edge of PLC strobe).
- i) AGV ready to read new order.

